



CHAI Shop & Co — CRM APPLICATION

System Architecture Diagram • Cloud-Native • Secure • Scalable • 2024

1. CLIENT LAYER



Mobile App

Customer App • iOS/Android
Order • Loyalty • Profile



POS Terminal

In-Store Billing • Staff Access
Offline Capable • Receipts



Admin Web

Management Dashboard • Reporting
User & Role Management

2. API GATEWAY LAYER

API Gateway

Authentication • Rate Limiting • Load Balancing • Request Routing • Security & Throttling



HTTPS/TLS • Cloud Hosting

3. BACKEND SERVICES



Auth Service

User Authentication •
RBAC • JWT Tokens



Billing Service

Invoices • Payments
Subscription Billing



Loyalty Service

Points • Rewards
Promotions & Offers



Inventory Service

Stock Management •
Products • Warehousing



Analytics Service

Metrics • Dashboards
• Insights & Reports

4. DATABASE LAYER



PostgreSQL

Primary Relational DB
Users • Orders • Products • Transactions



Redis Cache

Session Cache • Hot Data
Caching • TTL • Session Storage



Firebase Storage

Media & File Storage
Receipts • Images • Documents



Firegapse Storage

Media & File Storage
Receipts • Images • Documents



Swiggy / Zomato

Food Delivery Integration
Order Fulfillment • Partner APIs

5. EXTERNAL INTEGRATIONS



WhatsApp API

Customer Notifications • Messaging
Order Updates • Alerts



Razorpay Payment

Payment Gateway • UPI/Card/Wallet
Refunds • Transactions



Google Maps

Location Services • Geofencing
Delivery Tracking • Routing



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CHAI Shop & Co CRM — Design Template

Wireframe vs Final UI • 5 Core Screens

TOP ROW — WIREFRAMES (LOW-FIDELITY)

Grayscale • Boxes & Lines • Layout Structure



BOTTOM ROW — FINAL UI (HIGH-FIDELITY)

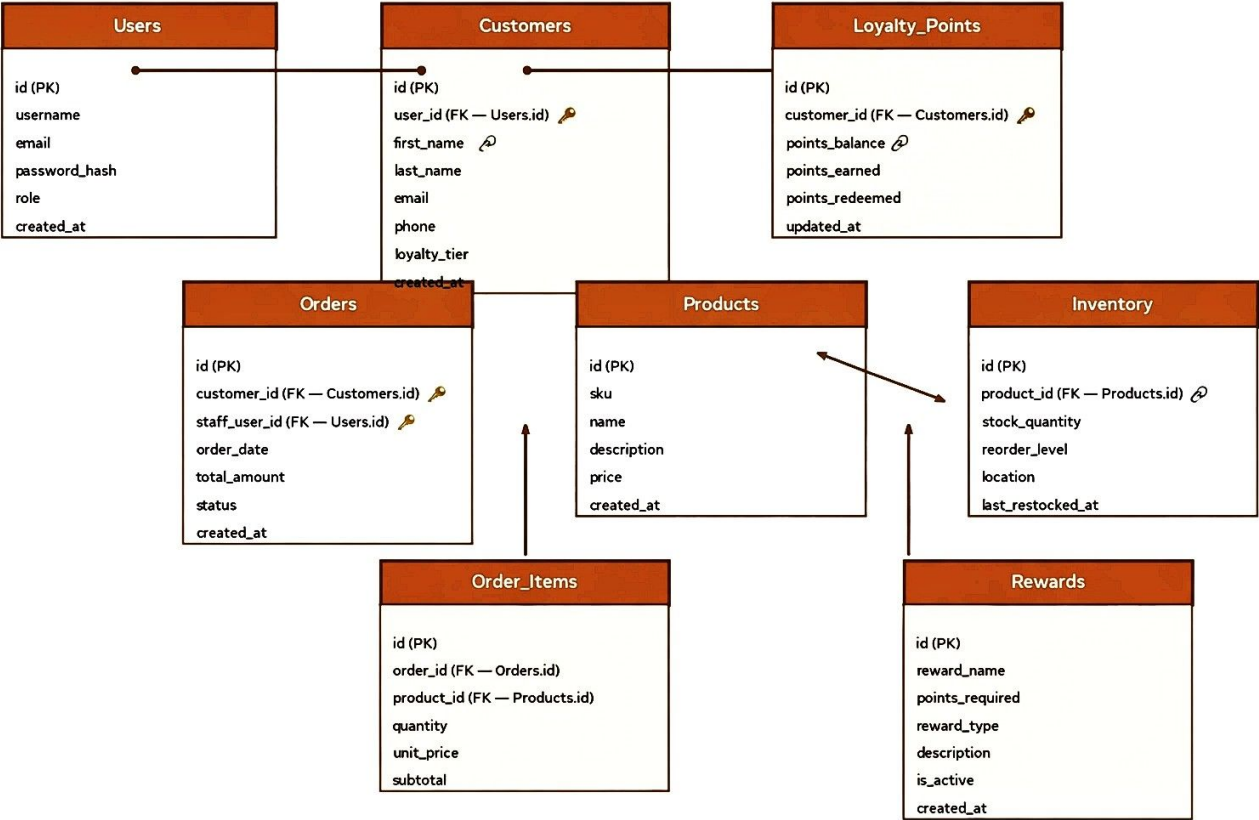
Warm Brown & Cream • Polished • Interactive UI



CHAI Shop & Co — CRM Database ER Diagram

Legend: PK = Primary Key FK = Foreign Key — = Relationship

Relationships: Customers 1—* Orders • Orders 1—* Order_Items • Order_Items *—1 Products • Products 1—1 Inventory • Customers 1—1 Loyalty_Points • Users 1—* Orders (staff)



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